

Vinay Reddy Oppuri

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Summary

Full Stack Engineer specializing in AI-integrated web applications and LLM-powered automation pipelines. Proficient in TypeScript, React, Next.js, and Node.js, with hands-on experience building RAG pipelines, LLM API integrations, and vector-based semantic search. Comfortable across the stack, from type-safe backend services to responsive, performant frontends, with working knowledge of core AWS services (EC2, S3).

Technical Skills

Languages: JavaScript, TypeScript, Python, C, C++

Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, shadcn/ui

Backend: Node.js, tRPC, REST API Design, FastAPI, Uvicorn, Pydantic

AI & GenAI: LLM APIs (Gemini), RAG Pipelines, Vector Embeddings, Hugging Face, Prompt Engineering

Databases: PostgreSQL (NeonDB), MongoDB, Supabase, Drizzle ORM

Cloud & DevOps: AWS (EC2, S3 - working knowledge), Vercel, Render, Docker, Git/GitHub, CI/CD, Turborepo

Auth & Security: Better Auth, JWT, Role-based Access Control

Experience

Website Development Team Member

IIT Raichur

Official Institute Websites — iiitr.ac.in

- Maintain and enhance multiple official web platforms under the `iiitr.ac.in` domain, driving modernization through a scalable **Next.js App Router** architecture.
- Built and shipped the **Training & Placement Cell Portal** (tnp-iiitr.vercel.app) end-to-end using **Next.js** and **Tailwind CSS**.
- Designed reusable, modular component libraries adopted across multiple institute projects, reducing duplicate code and speeding up development.

Projects

Rescomail [Next.js, TypeScript, Turborepo, Python, FastAPI, Gemini API, Trigger.dev v3, Tavily, Drizzle ORM, Better Auth, PostgreSQL, Vercel, Render]

- Architected a full-stack AI job copilot in a Turborepo monorepo, pairing a Next.js frontend with a FastAPI microservice deployed independently on Vercel and Render, connected via shared-secret authentication.
- Built an ATS intelligence pipeline using Gemini embeddings and cosine similarity for RAG-based skill-gap analysis and job-fit compatibility scoring.
- Automated cold email generation via Gemini with model failover, Pydantic schemas for structured output, and Tavily web scraping for real-time company context.
- Built a type-safe data layer with Drizzle ORM on PostgreSQL, integrating Better Auth, UploadThing, and a Trigger.dev (v3) background job queue for async processing.

Mono

[Next.js, TypeScript, Node.js, tRPC, PostgreSQL, Drizzle ORM, Better Auth, Gemini AI]

- Built an AI agent workspace enabling users to create task-specific LLM-powered assistants that automate repetitive business workflows.
- Integrated the Gemini LLM API to deliver context-aware responses grounded in saved conversation history, implementing a lightweight RAG-style retrieval pattern for recurring task automation.
- Engineered type-safe backend APIs with Node.js, tRPC, and Drizzle ORM on PostgreSQL, implementing user-scoped access control and secure session management via Better Auth.

Certifications

Cloud Computing - NPTEL (IIT Kharagpur), via SWAYAM

Jan-Apr 2026

Elite Certification - Consolidated Score: 78%

Education

in Artificial Intelligence & Data Science

B.Tech

Indian Institute of Information Technology, Raichur

Expected 2027

JEE Mains 2023: Ranked in top 4.5% nationally.